

General Chemistry 1 (CHEM 1141)

Shawnee State University – Fall 2019

December 5, 2019

Exam # 4 A

Name _____

*Please write your full name, and the exam version (4 A) that you have on the scantron sheet !
(Bubble in the best answer choice for each question on the green & white scantron sheet in pencil !)*

Please ☒ check the box next to your correct section number.

- | | | | |
|-------------------|---|---|----------------|
| Section #: | ☐ 1. (Monday Lab, 11:10 AM – 1:55 PM) | } | Fleeman |
| | ☐ 2. (Wednesday Lab, 11:10 AM – 1:55 PM) | | |
| | ☐ 3. (Monday Lab, 2:30 PM – 5:20 PM) | } | Napper |
| | ☐ 4. (Wednesday Lab, 2:30 PM – 5:20 PM) | | |
| | ☐ 5. (Thursday Lab, 12:30 PM – 3:20 PM) | } | Finnen |
| | ☐ 6. (Tuesday Lab, 12:30 PM – 3:20 PM) | | |

Multiple Choice: _____ / 50

Q21: _____ / 10

Q22: _____ / 10

Q23: _____ / 10

Q24: _____ / 10

Q25: _____ / 10

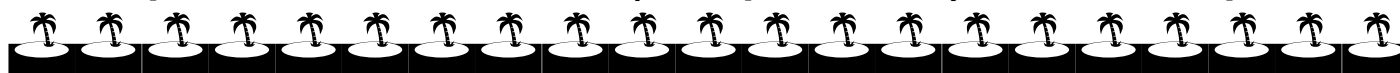
BONUS: _____ / 5

TOTAL: _____ / 100

You are only allowed to use a TI30–XIIS or equivalent non-programmable calculator on this exam !
(This means no cell phones, no smart phones, no smart watches, no iPads, or any other such devices will be allowed !)



Each problem in this section (multiple choice) is worth 2.5 points !

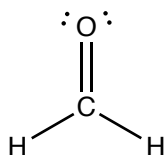


- Q1. Who is credited with the invention of the periodic table in the late 1860s, whereby the elements were ordered by atomic mass?
- A) Pauli
 - B) Schrodinger
 - C) Moseley
 - D) Mendeleyev
- Q2. Which of the following atoms will be **paramagnetic** in their ground state?
- A) helium
 - B) beryllium
 - C) neon
 - D) silicon
- Q3. The chemical equation corresponding to the first ionization energy of carbon is:
- A) $\text{C(g)} \rightarrow \text{C}^+(\text{g}) + \text{e}^-$
 - B) $\text{C(g)} + \text{e}^- \rightarrow \text{C}^-(\text{g})$
 - C) $\text{C}^+(\text{g}) \rightarrow \text{C}^{2+}(\text{g}) + \text{e}^-$
 - D) $\text{C}^{2+}(\text{g}) + \text{e}^- \rightarrow \text{C}^+(\text{g})$
- Q4. Which compound below has an expanded octet:
- A) BeCl_2
 - B) CCl_4
 - C) SCl_4
 - D) SiCl_4

Q5. The bond angle in a tetrahedral molecule is

- A) 90°
- B) 109.5°
- C) 120°
- D) 180°

Q6. How many sigma and pi bonds are there in the following molecule:



- A) 4 sigma, 0 pi
- B) 3 sigma, 1 pi
- C) 2 sigma, 2 pi
- D) 1 sigma, 3 pi

Q7. What is the correct name of the compound, Cr_2O_3

- A) chromium oxide
- B) dichromium trioxide
- C) chromium(II) oxide
- D) chromium(III) oxide

Q8. How many **atoms** are there in 16.04 g of CH_4 ?

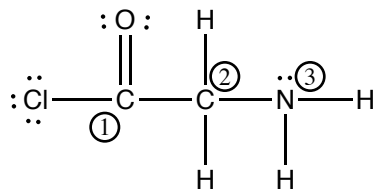
- A) 6.022×10^{23}
- B) 1.204×10^{23}
- C) 3.011×10^{24}
- D) 4.306×10^{22}

Q9. A hydrocarbon contains 81.71 % carbon and 18.29 % hydrogen by mass. Its empirical formula is:

- A) CH_2
- B) CH_3
- C) C_2H_9
- D) C_3H_8

- Q10. Which quantum number determines the **shape** of an orbital?
- A) n
 - B) l
 - C) m_l
 - D) m_s
- Q11. Hot tea is a solution containing caffeine, water, and various polyphenols that are present in a consistent composition throughout. It can best be described as being a(n):
- A) heterogeneous mixture
 - B) compound
 - C) homogeneous mixture
 - D) extensive solution
- Q12. Which of the following atoms or ions would have the **smallest** radius?
- A) K
 - B) K^+
 - C) Na
 - D) Na^+
- Q13. Which diatomic molecule would contain the **shortest** covalent bond?
- A) Br_2
 - B) Cl_2
 - C) O_2
 - D) N_2
- Q14. According to valence bond theory, which orbitals on bromine atoms overlap in the formation of the bond in Br_2 ?
- A) 4s
 - B) 4p
 - C) 4d
 - D) 4f

Q15. Consider the molecule below. Determine the hybridization at each of the three labeled atoms.



- A) 1 = sp^2 , 2 = sp^3 , 3 = sp^3
- B) 1 = sp^2 , 2 = sp^3 , 3 = sp^2
- C) 1 = sp^3 , 2 = sp^3 , 3 = sp^3
- D) 1 = sp^3 , 2 = sp^3 , 3 = sp^2

Q16. Choose the ground state electron configuration for Cr^{3+}

- A) $[Ar] 3d^3$
- B) $[Ar] 4s^1 3d^2$
- C) $[Ar] 4s^2 3d^6$
- D) $[Ar] 4s^2 3d^1$

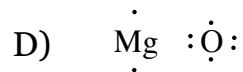
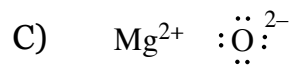
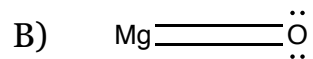
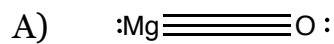
Q17. Give the molecular geometry and the number of *electron groups* (#repulsions) for BrF_5 .

- A) seesaw, 5 electron groups
- B) square pyramidal, 6 electron groups
- C) t-shaped, 5 electron groups
- D) octahedral, 6 electron groups

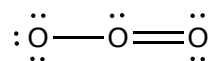
Q18. Choose the compound below that contains at least one polar covalent bond but is nonpolar.

- A) CF_4
- B) HCN
- C) $SeBr_4$
- D) ICl_3

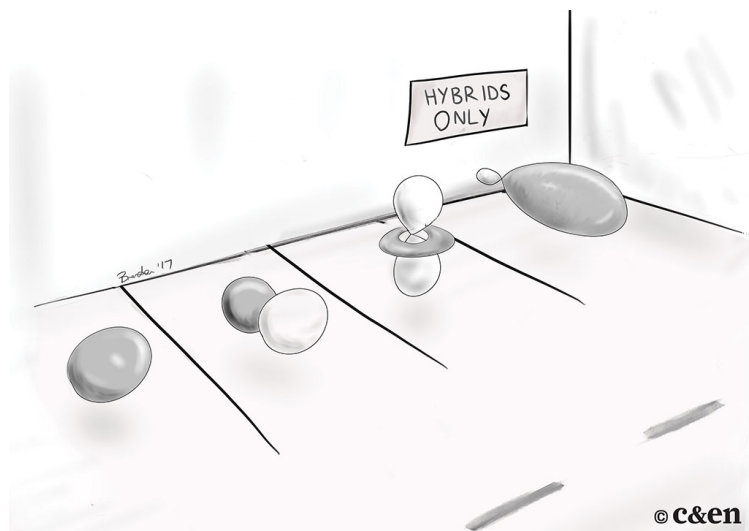
Q19. Identify the correct Lewis structure for MgO



Q20. What is the best explanation for the fact that both bonds in ozone, O_3 , have exactly the same strength?



- A) The central atom is sp^3 hybridized
- B) Both outer atoms have complete octets
- C) Ozone has resonance structures
- D) Ozone is a stable form of the element, oxygen





Each problem in this section (short answer) is worth 10 points !

All work must be show in order to receive credit !

You must use the factor–label (conversion–factor) method for all conversions !

Be sure to include units where applicable !

All numeric answers must be rounded to the correct number of significant figures !



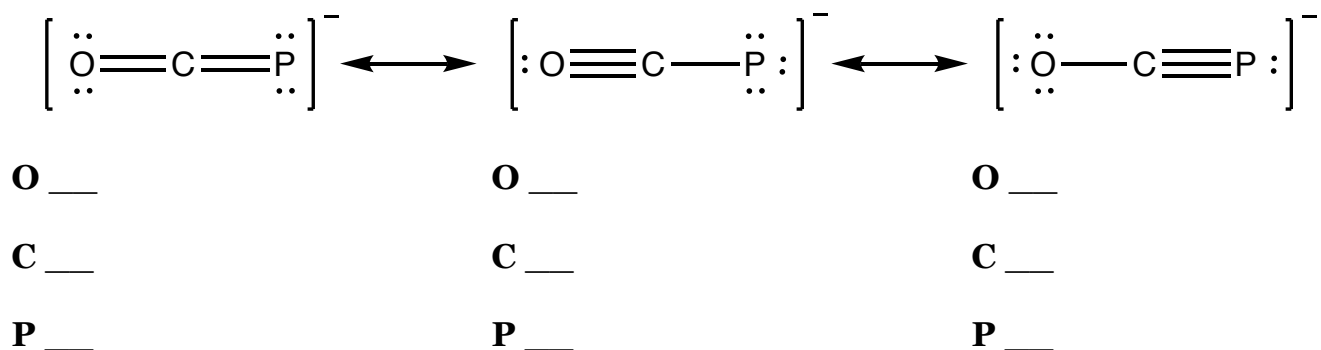
Q21. Place the correct number next to the letter that best matches.
(use each number only once)

- | | | |
|--------|--|----------------------------|
| ___ A. | Increases across the periods from left to right | 1. Beryllium |
| ___ B. | States that electrons enter unoccupied orbitals in a subshell with parallel spins before pairing up | 2. First ionization energy |
| ___ C. | An element that is capable of expanding its octet | 3. Auf Bau |
| ___ D. | An element that is frequently electron deficient | 4. Hund's rule |
| ___ E. | Decreases across the periods from left to right | 5. Shielding |
| ___ F. | States that electrons tend to enter lower energy subshells first before entering higher energy subshells | 6. Sulfur |
| ___ G. | The effect by which core electrons tend to reduce the effective nuclear charge felt by valence electrons | 7. Pauli principle |
| ___ H. | States that electrons must have a unique set of quantum numbers in an atom | 8. Core |
| ___ I. | The 2s electrons in a magnesium atom | 9. Valence |
| ___ J. | The 2p electrons in a nitrogen atom | 10. Atomic radii |

Q22. The polyatomic ion, OCP^- has several resonance structures that contribute to its electronic description. Three possible resonance structures are drawn below.

i) Determine the formal charge for each atom in the structures:

(be sure to show your work)



ii) **Explain** which structure (left, middle, right) likely contributes the most to our electronic description of the ion?

Q23. Write out **full** electron configurations for the following atoms/ions:

i) Ti

ii) Cu

iii) Ni^{2+}

Write out the orbital diagram for:

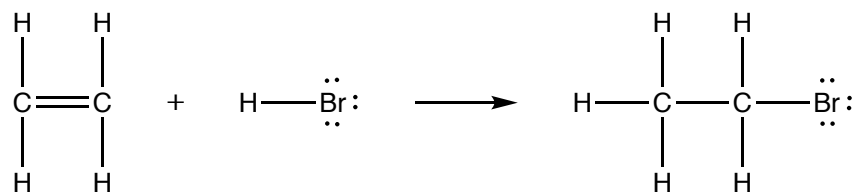
iv) Ti

Is Ti diamagnetic or paramagnetic? Explain!

Q24. Predict the molecular geometry and polarity of NF_3 . Your answer should include:

- ☐ A valid Lewis structure
- ☐ The total number of valence electrons
- ☐ A sketch of the geometry using line/dash/wedge notation
- ☐ The value of the bond angle(s) written out
- ☐ The name of both the **molecular** and **electron** geometry
- ☐ A clear explanation of why NF_3 is polar or non-polar

Q25. (i) Using the table of bond dissociation energies below, estimate ΔH for the following gas-phase chemical equation:



Bond	Bond enthalpy (kJ/mol)
C–C	348
C=C	614
C–H	413
H–Br	366
C–Br	276

(ii) What is the molecular geometry about each carbon atom in the **reactant** molecule, C_2H_4 ?

(iii) According to valence bond theory, the C–H bond in the **reactant** molecule, C_2H_4 forms from the overlap between which two orbitals?

_____ on Hydrogen, with _____ on Carbon

Exam checklist:

(Check the boxes to certify the following:)

- ☐ My full name is written legibly on the front page
- ☐ My correct lab section has been indicated on the front page
- ☐ My full name is written legibly on the scantron sheet
- ☐ My exam version (4A, 4B, 4C, or 4D) is written on the scantron sheet
- ☐ I have shown work for all problems (where appropriate), paying attention to
 - Significant figures / decimal places
 - Units
- ☐ I have used the conversion-factor method for all conversions

Thank-you from the Chemistry Professors and Good Luck!



Useful information:

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$$

[illegible]